



Universidad
del Caribe

2000

CANCUN, QUINTANA ROO, MÉXICO

CONOCIMIENTO Y CULTURA PARA EL DESARROLLO HUMANO

**Asignatura: Computo de alto
desempeño**

**Carrera: Ingeniería En Datos E
Inteligencia Organizacional.**

Profesor: Ismael Jiménez

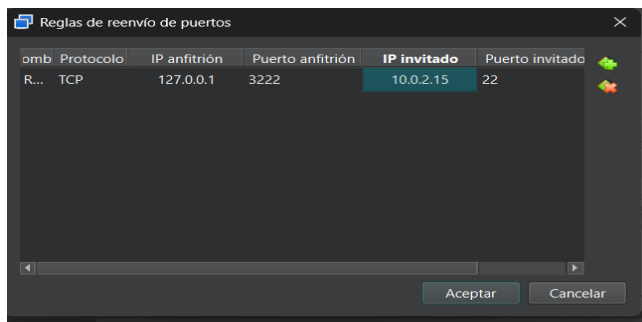
Actividad: Nodos

Fecha: 19/09/2025

**Alumna: Duran González Samantha
Camila**

Reporte de instalación: Galera 4 Cluster con MariaDB en Linux

Creación de máquinas virtuales con el SO Ubuntu 24.04.3 live-server Nodo 1



```
nodo1 login:
nodo1 login:
nodo1 login:
nodo1 login:
nodo1 login: samlove
Password:
Login incorrect
nodo1 login: samlove
Password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-79-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Mon Sep 15 02:52:50 AM UTC 2025

System load:          0.08
Usage of /:            5.3% of 48.91GB
Memory usage:         6%
Swap usage:           0%
Processes:            92
Users logged in:      0
IPv4 address for enp0s3: 10.0.2.15
IPv6 address for enp0s3: fd00::a00:27ff:fe2c:9830

Expanded Security Maintenance for Applications is not enabled.

10 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

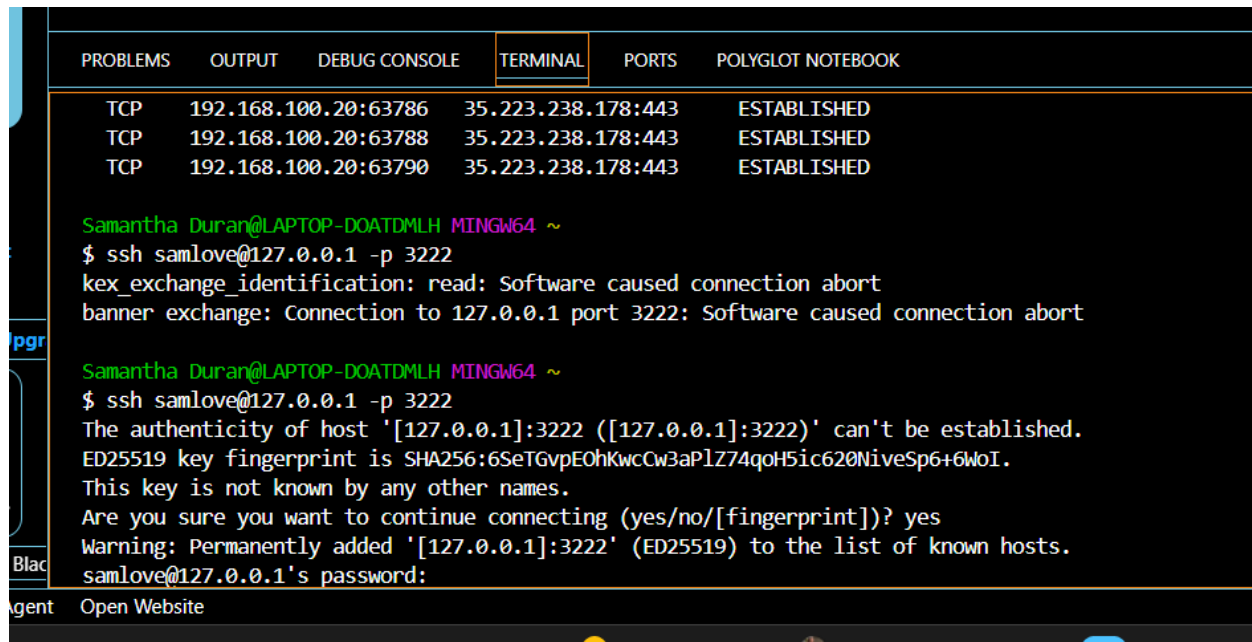
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
```

Esto nos va a permitir conectarnos vía SSH Se accede al nodo 1 e insertamos estos comandos:

Los cuales son la instalacion de paquetes y herramientas.

```
1 apt -y install net-tools
2 apt-get install software-properties-common
3 apt update
4 apt-get install mariadb-server mariadb-client galera-4
5 apt-get install galera-arbitrator-4
6 apt-get install mariadb-client libmariadb3
7 systemctl stop mysql
8 systemctl status mysql
9 vi /etc/mysql/mariadb.conf.d/60-galera.cnf
10 galera_new_cluster
11 mysql -u root -p -e "SHOW STATUS LIKE 'wsrep_cluster_size'"
12 mysql -u root --execute="SHOW GLOBAL STATUS WHERE Variable_name IN
('wsrep_ready', 'wsrep_cluster_size', 'wsrep_cluster_status',
'wsrep_connected');"
13 netstat -tlnp
14
15 apt install sysbench
16 mysql -uroot -p -e "create database sbtest"
17 sysbench --test=oltp --num-threads=16 --max-requests=10000
```

Ejecución con máquina virtual de visual o Git bash.



The screenshot shows a terminal window with a dark background. At the top, there is a header bar with tabs: PROBLEMS, OUTPUT, DEBUG CONSOLE, TERMINAL (which is active and highlighted), PORTS, and POLYGLOT NOTEBOOK. Below the header, the terminal displays three lines of network data:

PROBLEMS	OUTPUT	DEBUG CONSOLE	TERMINAL	PORTS	POLYGLOT NOTEBOOK
TCP	192.168.100.20:63786	35.223.238.178:443	ESTABLISHED		
TCP	192.168.100.20:63788	35.223.238.178:443	ESTABLISHED		
TCP	192.168.100.20:63790	35.223.238.178:443	ESTABLISHED		

Below the network data, the terminal shows the user 'Samantha Duran' at 'LAPTOP-DOATDMLH' running 'MINGW64'. The user attempts to connect via SSH to 'samlove@127.0.0.1' on port 3222. The first attempt fails with the message: 'kex_exchange_identification: read: Software caused connection abort' and 'banner exchange: Connection to 127.0.0.1 port 3222: Software caused connection abort'. The second attempt shows the host's authenticity, displaying a warning about a new key fingerprint and asking for confirmation to connect. The user responds 'yes', and the connection is established, showing the prompt 'samlove@127.0.0.1's password:'.

Comandos a ejecutar en el nodo 1:

```
apt -y install net-tools apt -y install software-properties-common
apt update apt -y
install mariadb-server
mariadb-client galera-4 apt install galera-arbitrator-4 apt
install mariadb-client libmariadb3
systemctl stop
mysql systemctl
status mysql
```

Configuramos el archivo /etc/mysql/mariadb.conf/60-galera.conf con el siguiente contenido

```
[mysqld]
binlog_format=ROW
default-storage-engine=innodb
innodb_autoinc_lock_mode=2
bind-address=0.0.0.0

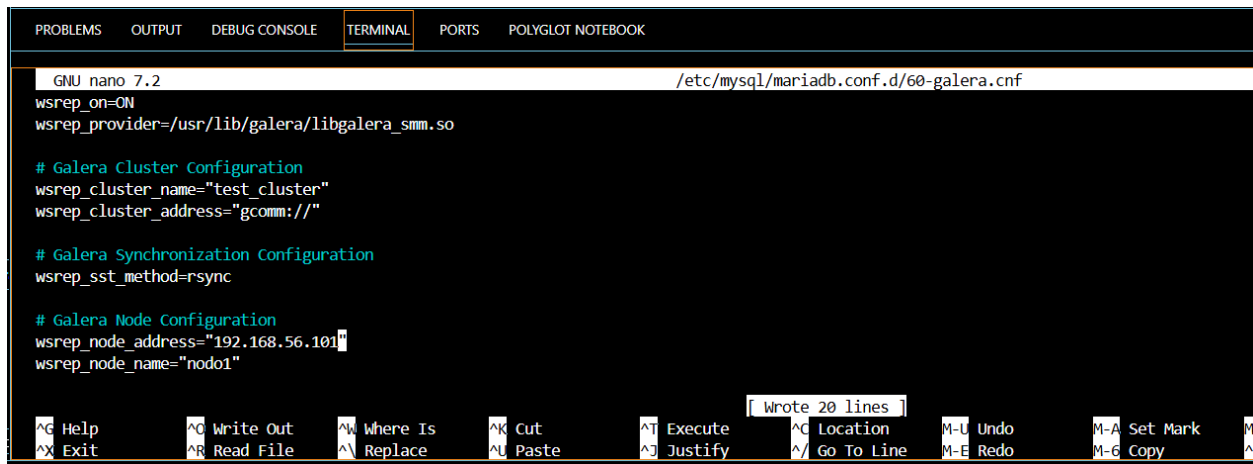
# Galera Provider Configuration
wsrep_on=ON
wsrep_provider=/usr/lib/galera/libgalera_smm.so

# Galera Cluster Configuration
wsrep_cluster_name="test_cluster"
wsrep_cluster_address="gcomm://192.168.56.101"

# Galera Synchronization Configuration
wsrep_sst_method=rsync

# Galera Node Configuration
wsrep_node_address="192.168.56.101"
```

wsrep_node_name="nodo1"
wsrep_node_address="192.168.56.101" wsrep_node_name="nodo1"



The screenshot shows a terminal window with the nano 7.2 editor open at the file /etc/mysql/mariadb.conf.d/60-galera.cnf. The configuration includes setting wsrep_on=ON, wsrep_provider=/usr/lib/galera/libgalera_smm.so, and defining a test cluster with three nodes. The node configuration for 'nodo1' is being edited, with wsrep_node_address set to '192.168.56.101' and wsrep_node_name set to 'nodo1'. The terminal has a dark background with syntax highlighting in green and blue. A status bar at the bottom indicates 'Wrote 20 lines' and lists various editor shortcuts.

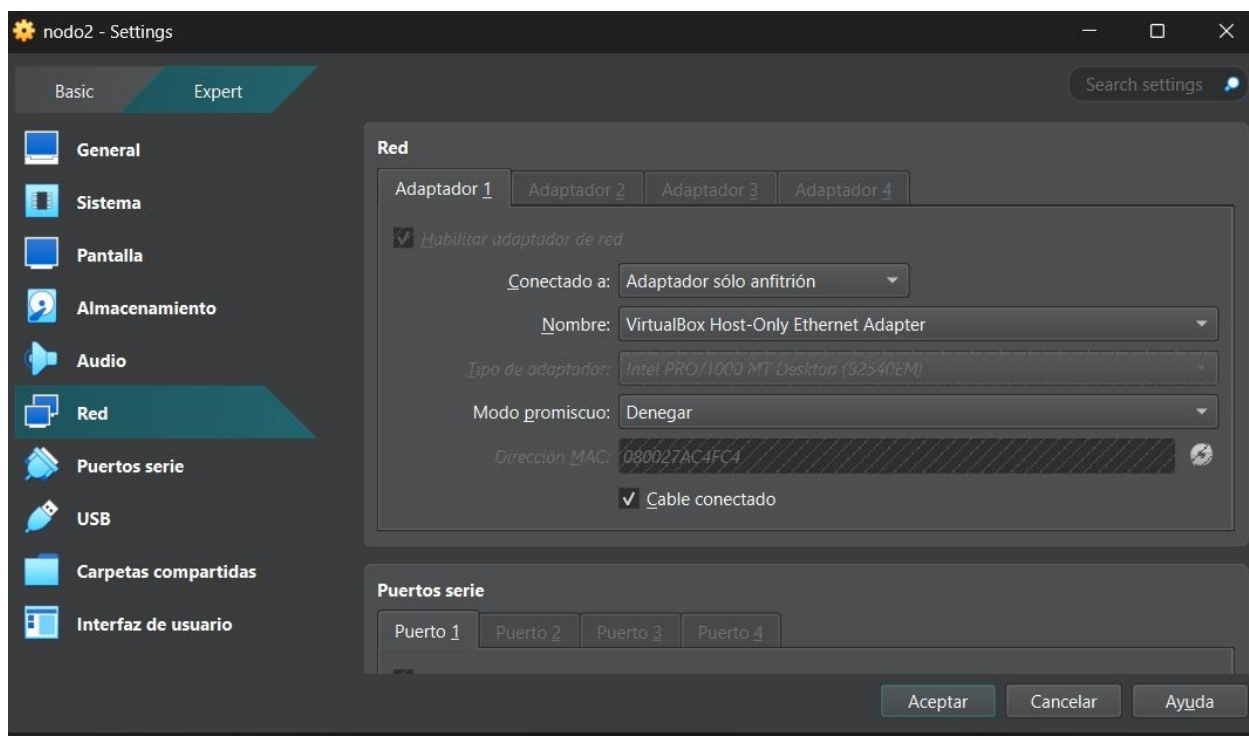
```
GNU nano 7.2 /etc/mysql/mariadb.conf.d/60-galera.cnf
wsrep_on=ON
wsrep_provider=/usr/lib/galera/libgalera_smm.so

# Galera Cluster Configuration
wsrep_cluster_name="test_cluster"
wsrep_cluster_address="gcomm://"

# Galera Synchronization Configuration
wsrep_sst_method=rsync

# Galera Node Configuration
wsrep_node_address="192.168.56.101"
wsrep_node_name="nodo1"
```

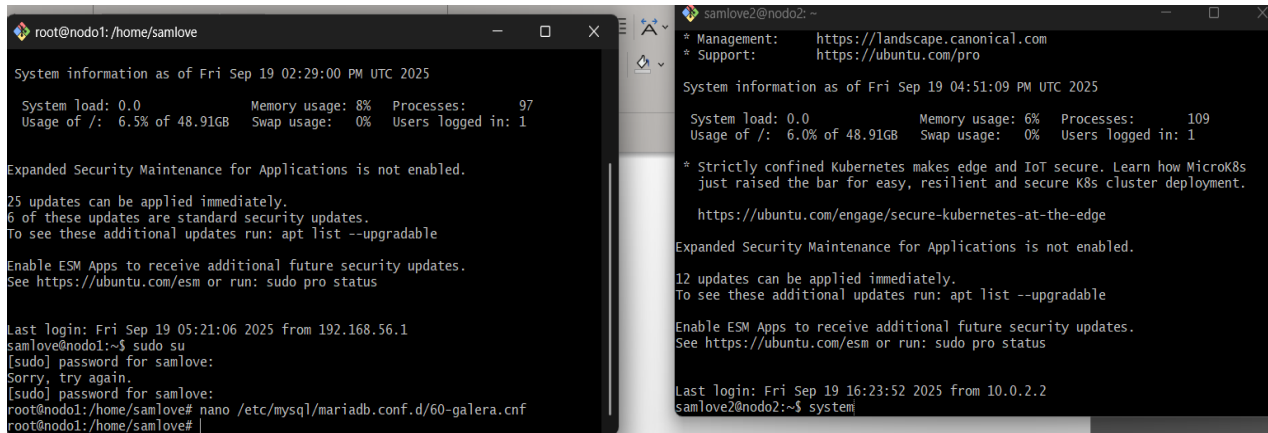
Cambiamos la configuración de la red



Corremos el comando galera_new_cluster

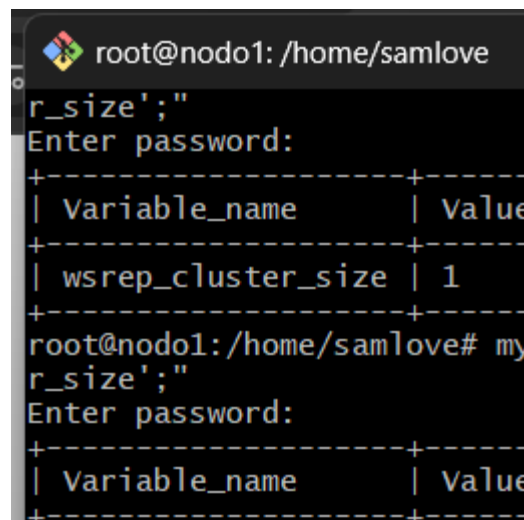
Ejecutamos los siguientes comandos: `mysql -u root -p -e "SHOW STATUS LIKE 'wsrep_cluster_size'"` y `mysql -u root --execute="SHOW GLOBAL STATUS WHERE Variable_name IN ('wsrep_ready', 'wsrep_cluster_size', 'wsrep_cluster_status', 'wsrep_connected');"`

Creo la base de datos y compruebo que se refleje con mi segundo nodo



The image shows two terminal windows side-by-side. The left window is titled 'root@nodo1: /home/samlove' and shows system information as of Fri Sep 19 02:29:00 PM UTC 2025. It lists system load, memory usage, processes, and updates. The right window is titled 'samlove2@nodo2: ~' and shows system information as of Fri Sep 19 04:51:09 PM UTC 2025. It also lists system load, memory usage, processes, and updates. Both windows show that expanded security maintenance for applications is not enabled and provide links to Ubuntu's security updates.

Instalación de sysbench en nodo1



The image shows a terminal window titled 'root@nodo1: /home/samlove'. It displays the execution of a MySQL command to check the cluster size. The command is `mysql -u root -p -e "SHOW STATUS LIKE 'wsrep_cluster_size'"`. The output shows a table with two columns: 'Variable_name' and 'Value'. The value for 'wsrep_cluster_size' is 1. The user then enters the password and the command is executed again, showing the same output.

Inserto 1000 datos dummy a la base de datos

```

root@nodo1: /home/samlove
Inserting 10000 records into 'sbtest5'
Creating a secondary index on 'sbtest5'...
root@nodo1:/home/samlove# sysbench oltp_read_write \
--mysql-host=127.0.0.1 \
--mysql-user=bench \
--mysql-password=bench123 \
--mysql-db=sbtest \
--tables=5 \
--table-size=10000 \
--threads=1 \
--time=60 \
run
sysbench 1.0.20 (using system LuaJIT 2.1.0-beta3)

Running the test with following options:
Number of threads: 1
Initializing random number generator from current time

Initializing worker threads...

Threads started!

```

```

root@nodo1:/home/samlove
Initializing worker threads...

Threads started!

SQL statistics:
  queries performed:
    read:          14812
    write:         3363
    other:         2985
    total:        21160
  transactions:    1058 (17.62 per sec.)
  queries:         21160 (352.46 per sec.)
  ignored errors:  0 (0.00 per sec.)
  reconnects:      0 (0.00 per sec.)

General statistics:
  total time:      60.0263s
  total number of events: 1058

Latency (ms):
  min:             26.37
  avg:             56.72
  max:             172.89
  95th percentile: 78.60

```

Reviso el rendimiento del nodo1

```

top - 15:26:23 up 16 min, 1 user, load average: 0.00, 0.04, 0.09
Tasks: 99 total, 1 running, 98 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 5.2 sy, 0.0 ni, 78.5 id, 0.0 wa, 0.0 hi, 16.3 si, 0.0 st
MiB Mem : 1968.2 total, 1438.7 free, 412.2 used, 263.6 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 1556.0 avail Mem

```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1016	jhair	20	0	15124	7108	5120	S	4.7	0.4	0:03.19	sshd
132	root	20	0	0	0	0	I	0.7	0.0	0:08.37	kworker/0:3-ev+
16	root	20	0	0	0	0	S	0.3	0.0	0:00.45	ksoftirqd/0
1176	mysql	20	0	1295996	148348	30848	S	0.3	7.4	0:03.13	mariadb
1	root	20	0	22180	13300	9460	S	0.0	0.7	0:01.80	systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kthreadd
3	root	20	0	0	0	0	S	0.0	0.0	0:00.00	pool_workqueue+
4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-rcu_g
5	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-rcu_p
6	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-slab_
7	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-netns
9	root	20	0	0	0	0	I	0.0	0.0	0:00.21	kworker/0:1-cg+
11	root	20	0	0	0	0	I	0.0	0.0	0:00.00	kworker/u2:0-f+
12	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-mm_pe
13	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_kthr+
14	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_rude+
15	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_trac+
17	root	20	0	0	0	0	I	0.0	0.0	0:00.12	rcu_preempt
18	root	rt	0	0	0	0	S	0.0	0.0	0:00.01	migration/0
19	root	-51	0	0	0	0	S	0.0	0.0	0:00.00	idle_inject/0
20	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/0

Se generan eventos de tipo read en 60 segundos y vemos sus resultados:

```
root@nodo1:/home/jhair# sysbench --threads=1 --time=60 --rate=0 --db-driver=mysql --mysql-user=root --events=0 o1tp_read_only run
sysbench 1.0.20 (using system LuaJIT 2.1.0-beta3)

Running the test with following options:
Number of threads: 1
Initializing random number generator from current time

Initializing worker threads...

Threads started!

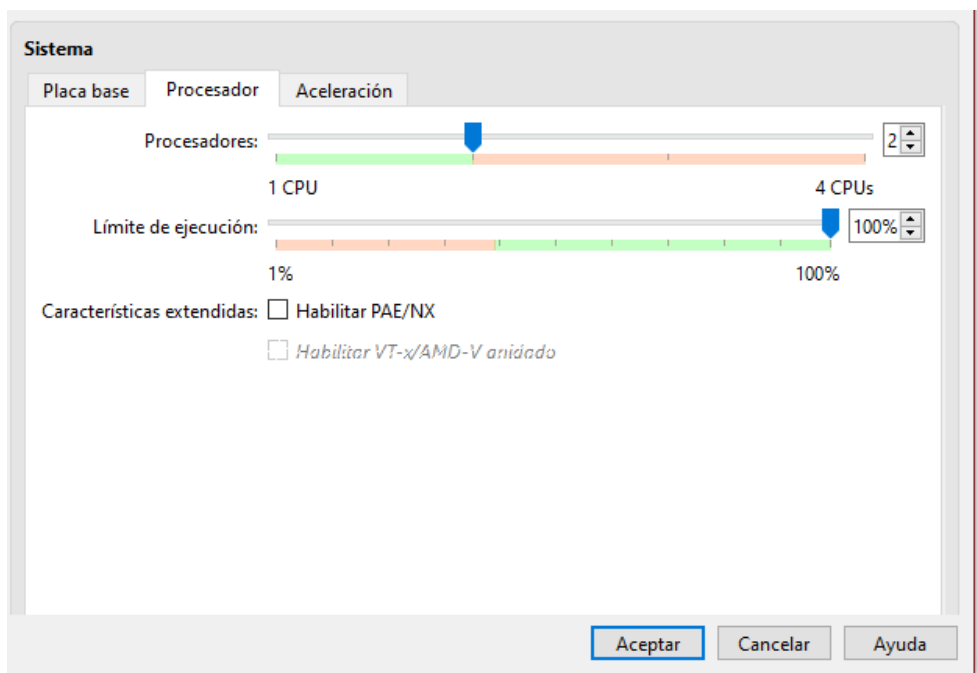
SQL statistics:
  queries performed:
    read:                355670
    write:                 0
    other:               50810
    total:             406480
  transactions:        25405 (423.38 per sec.)
  queries:             406480 (6774.02 per sec.)
  ignored errors:       0 (0.00 per sec.)
  reconnects:          0 (0.00 per sec.)

General statistics:
  total time:           60.0037s
  total number of events: 25405

Latency (ms):
  min:                   1.21
  avg:                   2.36
  max:                   85.17
  95th percentile:      5.77
  sum:                  59874.75

Threads fairness:
  events (avg/stddev):   25405.0000/0.00
  execution time (avg/stddev): 59.8748/0.00
```

Se actualiza a dos CPU para el nodo1



Cluster con dos nodos:

```
root@nodo3: /home/samlove3
GNU nano 7.2 /etc/mysql/mariadb.conf.d/60-galera.cnf *
binlog_format=ROW
default-storage-engine=innodb
innodb_autoinc_lock_mode=2
bind-address=0.0.0.0

# Galera Provider Configuration
wsrep_on=ON
wsrep_provider=/usr/lib/galera/libgalera_smm.so

# Galera Cluster Configuration
wsrep_cluster_name="test_cluster"
wsrep_cluster_address="gcomm://192.168.56.101,192.168.56.103,192.168.56.104"

# Galera Synchronization Configuration
wsrep_sst_method=rsync

# Galera Node Configuration
wsrep_node_address="192.168.56.104"
wsrep_node_name="nodo3"

[Help] [Write Out] [Where Is] [Cut] [Execute] [Location]
[Exit] [Read File] [Replace] [Paste] [Justify] [Go To Line]
```

```
root@nodo1: /home/samlove
25 updates can be applied immediately.
6 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your
Internet connection or proxy settings

Last login: Sat Sep 20 07:22:24 2025 from 192.168.56.1
samlove@nodo1:~$ sudo su
[sudo] password for samlove:
root@nodo1: /home/samlove# nano /etc/mysql/mariadb.conf.d/60-galera.cnf
root@nodo1: /home/samlove# galera_new_cluster
root@nodo1: /home/samlove# mysql -uroot -e "SHOW STATUS LIKE 'wsrep_cluster_size'"
+-----+-----+
| Variable_name | Value |
+-----+-----+
| wsrep_cluster_size | 2 |
+-----+-----+
root@nodo1: /home/samlove#
```

```
root@nodo2p: /home/samlove2
● mariadb.service - MariaDB 10.11.13 database server
   Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; preset: >
   Active: active (running) since Sun 2025-09-21 05:12:06 UTC; 3min 3s ago
     Docs: man:mariabdb(8)
           https://mariadb.com/kb/en/library/systemd/
   Process: 3344 ExecStartPre=/usr/bin/install -m 755 -o mysql -g root -d /var>
   Process: 3346 ExecStartPre=/bin/sh -c systemctl unset-environment _WSREP_ST>
   Process: 3348 ExecStartPre=/bin/sh -c [ ! -e /usr/bin/galera_recovery ] &&>
   Process: 3715 ExecStartPost=/bin/sh -c systemctl unset-environment _WSREP_S>
   Process: 3718 ExecStartPost=/etc/mysql/debian-start (code=exited, status=0/>
   Main PID: 3456 (mariabdb)
   Status: "Taking your SQL requests now..."
     Tasks: 15 (limit: 29639)
    Memory: 218.3M (peak: 221.4M)
       CPU: 12.178s
    CGroup: /system.slice/mariadb.service
            └─3456 /usr/sbin/mariabdb --wsrep_start_position=00000000-0000-000>

Sep 21 05:12:06 nodo2p mariabdb[3456]: 2025-09-21 5:12:06 0 [Note] WSREP: Proc>
Sep 21 05:12:06 nodo2p mariabdb[3456]: 2025-09-21 5:12:06 0 [Note] WSREP: Shif>
Sep 21 05:12:06 nodo2p mariabdb[3456]: 2025-09-21 5:12:06 2 [Note] WSREP: Serv>
Sep 21 05:12:06 nodo2p mariabdb[3456]: 2025-09-21 5:12:06 2 [Note] WSREP: Serv>
Sep 21 05:12:06 nodo2p mariabdb[3456]: 2025-09-21 5:12:06 2 [Note] WSREP: Sync>
root@nodo2p: /home/samlove2#
```


Realización de pruebas para las diferentes tareas

```
root@nodo1: /home/samlove
--- oltp_point_select con 2 hilo(s) ---
transactions: 558342 (9305.16 per sec.)
>>> Ejecutando prueba: oltp_read_only
--- oltp_read_only con 1 hilo(s) ---
--- oltp_read_only con 2 hilo(s) ---
>>> Ejecutando prueba: oltp_read_write
--- oltp_read_write con 1 hilo(s) ---
--- oltp_read_write con 2 hilo(s) ---
>>> Ejecutando prueba: oltp_update_index
--- oltp_update_index con 1 hilo(s) ---
--- oltp_update_index con 2 hilo(s) ---
>>> Ejecutando prueba: oltp_update_non_index
--- oltp_update_non_index con 1 hilo(s) ---
--- oltp_update_non_index con 2 hilo(s) ---
>>> Ejecutando prueba: oltp_write_only
--- oltp_write_only con 1 hilo(s) ---
--- oltp_write_only con 2 hilo(s) ---
>>> Ejecutando prueba: select_random_points
--- select_random_points con 1 hilo(s) ---
transactions: 8789 (146.47 per sec.)
--- select_random_points con 2 hilo(s) ---
transactions: 341403 (5689.15 per sec.)
>>> Ejecutando prueba: select_random_ranges
--- select_random_ranges con 1 hilo(s) ---
transactions: 11629 (193.79 per sec.)
--- select_random_ranges con 2 hilo(s) ---
transactions: 334631 (5576.34 per sec.)
root@nodo1: /home/samlove# cat resultados_benchmark.txt
=== Resultados de Benchmark Galera (2 nodos) ===
Usuario: bench | DB: sbtest | Tiempo: 60s
--- select_random_ranges con 1 hilo(s) ---
transactions: 11629 (193.79 per sec.)
--- select_random_ranges con 2 hilo(s) ---
transactions: 334631 (5576.34 per sec.)
benchmark completado. Resultados guardados en resultados_benchmark.txt
root@nodo1: /home/samlove# cat resultados_benchmark.txt
=== Resultados de Benchmark Galera (2 nodos) ===
Usuario: bench | DB: sbtest | Tiempo: 60s
>> Ejecutando prueba: bulk_insert
--- bulk_insert con 1 hilo(s) ---
--- bulk_insert con 2 hilo(s) ---
>> Ejecutando prueba: oltp_delete
--- oltp_delete con 1 hilo(s) ---
transactions: 15092 (251.48 per sec.)
--- oltp_delete con 2 hilo(s) ---
transactions: 67450 (1124.05 per sec.)
>> Ejecutando prueba: oltp_insert
--- oltp_insert con 1 hilo(s) ---
transactions: 3379 (56.29 per sec.)
--- oltp_insert con 2 hilo(s) ---
transactions: 5326 (88.73 per sec.)
>> Ejecutando prueba: oltp_point_select
--- oltp_point_select con 1 hilo(s) ---
transactions: 39731 (661.99 per sec.)
--- oltp_point_select con 2 hilo(s) ---
transactions: 558342 (9305.16 per sec.)
>> Ejecutando prueba: oltp_read_only
--- oltp_read_only con 1 hilo(s) ---
--- oltp_read_only con 2 hilo(s) ---
>> Ejecutando prueba: oltp_read_write
--- oltp_read_write con 1 hilo(s) ---
--- oltp_read_write con 2 hilo(s) ---
p_update_index
1 hilo(s) ---
2 hilo(s) ---
p_update_non_index
root@nodo1: /home/samlove
>>> Ejecutando prueba: bulk_insert
--- bulk_insert con 1 hilo(s) ---
--- bulk_insert con 2 hilo(s) ---
>>> Ejecutando prueba: oltp_read_write
--- oltp_read_write con 1 hilo(s) ---
--- oltp_read_write con 2 hilo(s) ---
>>> Ejecutando prueba: oltp_update_index
--- oltp_update_index con 1 hilo(s) ---
--- oltp_update_index con 2 hilo(s) ---
>>> Ejecutando prueba: oltp_update_non_index
--- oltp_update_non_index con 1 hilo(s) ---
--- oltp_update_non_index con 2 hilo(s) ---
>>> Ejecutando prueba: oltp_write_only
--- oltp_write_only con 1 hilo(s) ---
--- oltp_write_only con 2 hilo(s) ---
>>> Ejecutando prueba: select_random_points
--- select_random_points con 1 hilo(s) ---
transactions: 8789 (146.47 per sec.)
--- select_random_points con 2 hilo(s) ---
transactions: 341403 (5689.15 per sec.)
>>> Ejecutando prueba: select_random_ranges
--- select_random_ranges con 1 hilo(s) ---
transactions: 11629 (193.79 per sec.)
--- select_random_ranges con 2 hilo(s) ---
transactions: 334631 (5576.34 per sec.)
root@nodo1: /home/samlove# sysbench /usr/share/sysbench/oltp_read_write.lua \
--mysql-user=bench \
--mysql-password=bench123 \
--mysql-db=sbtest \
--mysql-host=192.168.56.101 \
--tables=10 \
--table-size=10000 \
--threads=2 \
--time=60 \
run
sysbench 1.0.20 (using system LuaJIT 2.1.0-beta3)
Running the test with following options:
Number of threads: 2
Initializing random number generator from current time
```

Creación del tercer nodo:

```
root@nodo3: /home/samlove3
GNU nano 7.2 /etc/mysql/mariadb.conf.d/60-galera.cnf *
binlog_format=ROW
default-storage-engine=innodb
innodb_autoinc_lock_mode=2
bind-address=0.0.0.0

# Galera Provider Configuration
wsrep_on=ON
wsrep_provider=/usr/lib/galera/libgalera_smm.so

# Galera Cluster Configuration
wsrep_cluster_name="test_cluster"
wsrep_cluster_address="gcomm://192.168.56.101,192.168.56.103,192.168.56.104"

# Galera Synchronization Configuration
wsrep_sst_method=rsync

# Galera Node Configuration
wsrep_node_address="192.168.56.104"
wsrep_node_name="nodo3"

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute  ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify  ^_ Go To Line
```

Se hace la conexión con el tercer nodo para realizar las últimas pruebas

```
root@nodo1: /home/samlove
_size';"
Enter password:
-----+-----+
Variable_name | Value |
-----+-----+
wsrep_cluster_size | 1 |
-----+-----+
root@nodo1:/home/samlove# mysql -uroot -p -e "SHOW STATUS LIKE 'wsrep_cluste
_size';"
Enter password:
-----+-----+
Variable_name | Value |
-----+-----+
wsrep_cluster_size | 2 |
-----+-----+
root@nodo1:/home/samlove# mysql -uroot -p -e "SHOW STATUS LIKE 'wsrep_cluste
_size';"
Enter password:
-----+-----+
Variable_name | Value |
-----+-----+
wsrep_cluster_size | 3 |
-----+-----+
root@nodo1:/home/samlove#
```

```
root@nodo3: /home/samlove3
root@nodo3:/home/samlove3# mysql -uroot -p -e "SHOW STATUS LIKE 'wsrep_cluster_size';"
Enter password:
+-----+-----+
| Variable_name | Value |
+-----+-----+
| wsrep_cluster_size | 3 |
+-----+-----+
root@nodo3:/home/samlove3#
```

```
root@nodo2p: /home/samlove2
samlove2@nodo2p:~$ sudo su
[sudo] password for samlove2:
root@nodo2p:/home/samlove2# systemctl start mariadb
root@nodo2p:/home/samlove2# systemctl start mariadb
root@nodo2p:/home/samlove2# systemctl start mariadb
root@nodo2p:/home/samlove2# mysql -uroot -p -e "SHOW STATUS LIKE 'wsrep_cluster_size';"
Enter password:
+-----+-----+
| Variable_name | Value |
+-----+-----+
| wsrep_cluster_size | 2 |
+-----+-----+
root@nodo2p:/home/samlove2# systemctl stop mariadb
root@nodo2p:/home/samlove2# systemctl start mariadb
root@nodo2p:/home/samlove2# mysql -uroot -p -e "SHOW STATUS LIKE 'wsrep_cluster_size';"
Enter password:
+-----+-----+
| Variable_name | Value |
+-----+-----+
| wsrep_cluster_size | 3 |
+-----+-----+
root@nodo2p:/home/samlove2#
```

Pruebas con los tres nodos

```
root@nodo1: /home/samlove
--- oltp_write_only con 1 hilo(s) ---
--- oltp_write_only con 2 hilo(s) ---
--- select_random_points con 1 hilo(s) ---
transactions: 4437 (73.92 per sec.)
--- select_random_points con 2 hilo(s) ---
transactions: 27496 (458.10 per sec.)
--- select_random_ranges con 1 hilo(s) ---
transactions: 9373 (156.20 per sec.)
--- select_random_ranges con 2 hilo(s) ---
transactions: 27566 (459.36 per sec.)
>>> Ejecutando pruebas en nodo 192.168.56.104
--- oltp_delete con 1 hilo(s) ---
--- oltp_delete con 2 hilo(s) ---
--- oltp_insert con 1 hilo(s) ---
--- oltp_insert con 2 hilo(s) ---
root@nodo1: /home/samlove
```

```
root@nodo1: /home/samlove

-- oltp_write_only con 1 hilo(s) ---
-- oltp_write_only con 2 hilo(s) ---
-- select_random_points con 1 hilo(s) ---
  transactions:                16194  (269.89 per sec.)
-- select_random_points con 2 hilo(s) ---
  transactions:                405061 (6749.61 per sec.)
-- select_random_ranges con 1 hilo(s) ---
  transactions:                21443  (357.35 per sec.)
-- select_random_ranges con 2 hilo(s) ---
  transactions:                436681 (7276.10 per sec.)
>> Ejecutando pruebas en nodo 192.168.56.103
-- oltp_delete con 1 hilo(s) ---
-- oltp_delete con 2 hilo(s) ---
-- oltp_insert con 1 hilo(s) ---
```

```
root@nodo1: /home/samlove

--- oltp_update_index con 2 hilo(s) ---
--- oltp_update_non_index con 1 hilo(s) ---
--- oltp_update_non_index con 2 hilo(s) ---
--- oltp_write_only con 1 hilo(s) ---
--- oltp_write_only con 2 hilo(s) ---
--- select_random_points con 1 hilo(s) ---
  transactions:                3775   (62.90 per sec.)
--- select_random_points con 2 hilo(s) ---
  transactions:                26943  (448.93 per sec.)
--- select_random_ranges con 1 hilo(s) ---
  transactions:                9344   (155.71 per sec.)
--- select_random_ranges con 2 hilo(s) ---
  transactions:                26618  (443.52 per sec.)
root@nodo1: /home/samlove#
```

Interpretación de los resultados

Sincronización del clúster: El clúster de 3 nodos está funcionando correctamente, ya que las pruebas se pudieron ejecutar en cada nodo (192.168.56.101, 192.168.56.103 y 192.168.56.104). Los resultados muestran transacciones registradas en cada nodo, lo que confirma que la base de datos se replica y los nodos responden a las pruebas de carga. Pruebas con carga ligera (1 hilo):

Con un solo hilo, las tasas de transacciones por segundo son relativamente bajas (ej. ~62 a ~270 TPS en `select_random_points`). Esto refleja que la capacidad de procesamiento está limitada por la baja concurrencia, lo cual es esperado.

Pruebas con mayor concurrencia (2 hilos): Al aumentar a 2 hilos, el rendimiento crece de forma significativa. Ejemplo:

Nodo 192.168.56.101 en `select_random_points`: de 269 TPS a 6749 TPS.

Nodo 192.168.56.103 en `select_random_points`: de 73 TPS a 458 TPS.

Esto indica que el clúster responde bien a cargas concurrentes y escala con más hilos.

Comparación entre nodos: No todos los nodos muestran el mismo nivel de rendimiento. El nodo 192.168.56.101 tiene un rendimiento muy superior en varias pruebas, mientras que 192.168.56.103 y 192.168.56.104 presentan cifras más moderadas.

Esto puede deberse a:

- Diferencias en recursos asignados (CPU/RAM) en cada máquina virtual.
- Latencias de red entre los nodos.
- Sobrecarga por ser el nodo en el que corres directamente las pruebas (`sysbench` en 192.168.56.101).

Conclusión:

El clúster Galera con 3 nodos funciona correctamente y mantiene la replicación de datos. Los benchmarks con 2 hilos muestran una mejora clara en el rendimiento respecto a 1 hilo, lo que confirma que el sistema aprovecha la concurrencia. Sin embargo, se observan desbalances en el rendimiento entre los nodos, lo que sugiere que el nodo donde se ejecutan directamente las pruebas obtiene mayor ventaja. En general, demuestra que Galera replica correctamente, escala con la concurrencia y mantiene la consistencia entre los 3 nodos, cumpliendo con los objetivos.